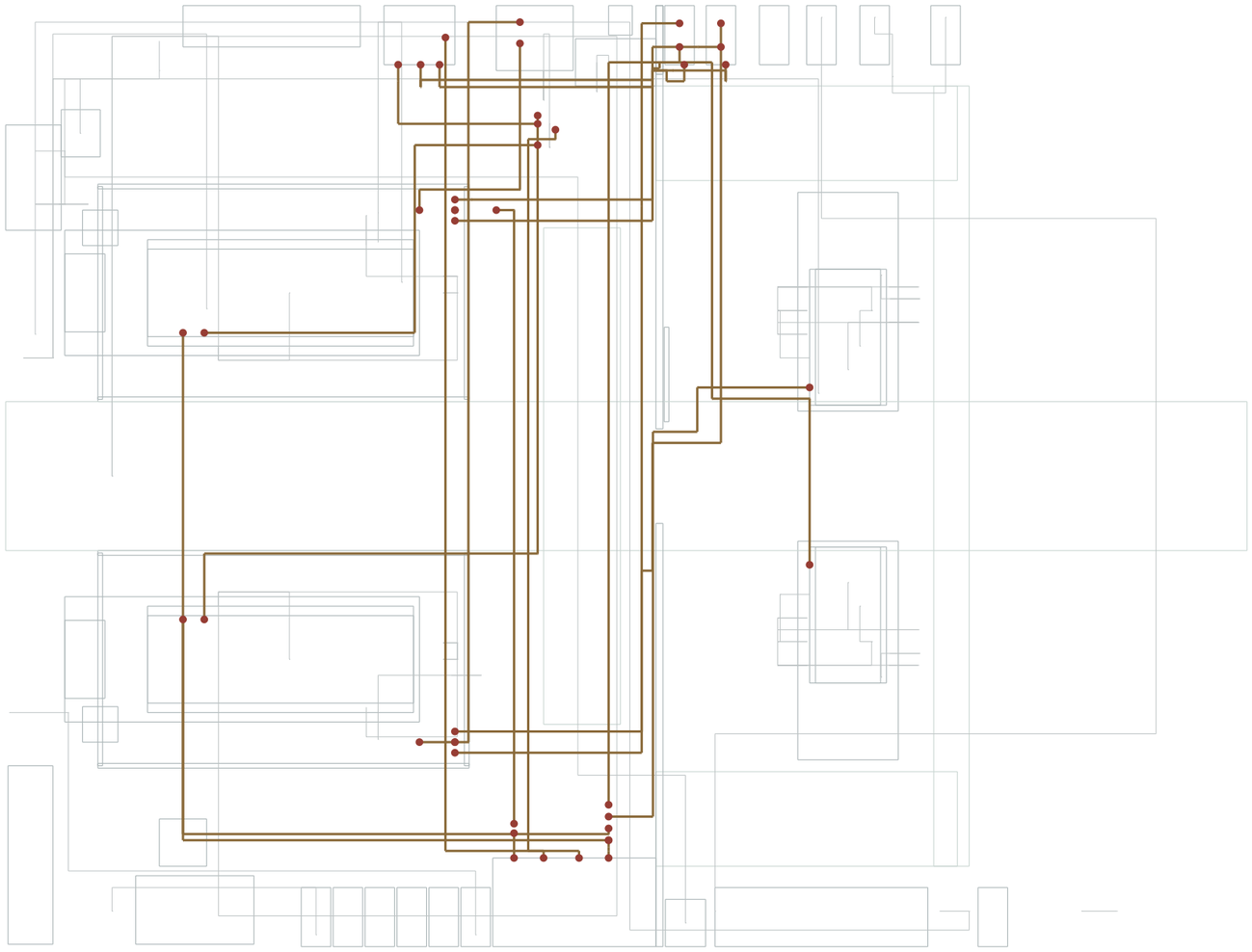
JS127-FUEL-C1-01 | Fuel external routes - plan

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED



5 m calibration bar: 100 mm at 1:50

ROUTE KEY

01 bunker > storeP DN50

02 bunker > storeS DN50

03 storeP > vP DN25

04 storeS > vS DN25

05 vP > tp1 DN25

06 vS > tp2 DN25

07 vP > tp2 DN25

08 vS > tp1 DN25

09 tp1 > purifier DN25

10 tp2 > purifier DN25

11 purifier > service DN25

12 purifier > sludge DN20

13 service > overflow DN32

14 overflow > storeP DN32

15 service > qcv DN20

16 qcv > fuel-mainP DN20

17 qcv > fuel-mainS DN20

18 qcv > fuel-gen1 DN15

19 qcv > fuel-gen2 DN15

20 fuel-mainP > return DN15

21 fuel-mainS > return DN15

22 return > service DN15

Brown: revised fuel routes. Grey: accepted source services. Pale green: proposed access reservations. Red dots: proposed ports; see terminal schedule.

Sharp corners show clearance envelopes only. Bend/flange/support design remains open. Tank partitions and internal manifold assemblies are not fabricated geometry.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

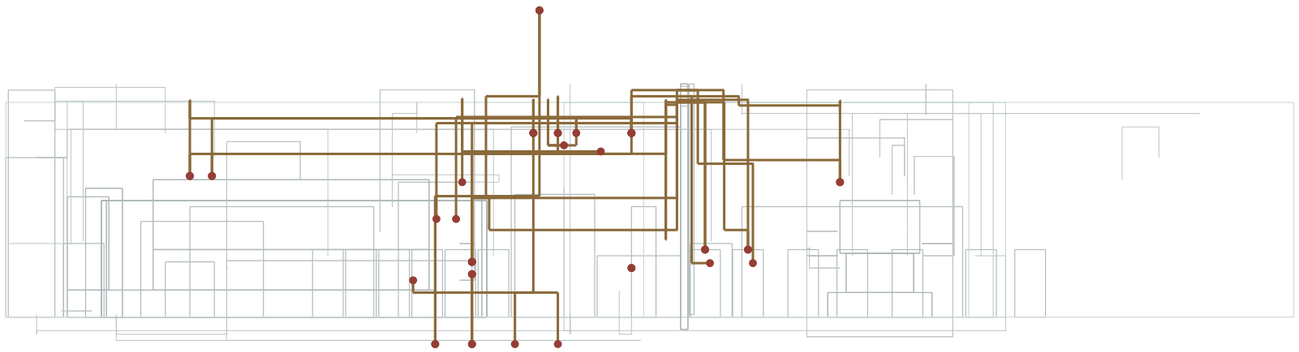
REV C1 | 2026-10-17 | FUEL-ONLY ADDENDUM | A3 | SCALE 1:50

DRAWN: generated | CHECK: internal geometric checks | APPROVAL: NONE

Reference: immutable Revision B (13 October 2026). Proposed interfaces; no supplier or construction approval.

JS127-FUEL-C1-02 | Fuel external routes - profile

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED



5 m calibration bar: 100 mm at 1:50

ROUTE KEY

01 bunker > storeP DN50

02 bunker > storeS DN50

03 storeP > vP DN25

04 storeS > vS DN25

05 vP > tp1 DN25

06 vS > tp2 DN25

07 vP > tp2 DN25

08 vS > tp1 DN25

09 tp1 > purifier DN25

10 tp2 > purifier DN25

11 purifier > service DN25

12 purifier > sludge DN20

13 service > overflow DN32

14 overflow > storeP DN32

15 service > qcv DN20

16 qcv > fuel-mainP DN20

17 qcv > fuel-mainS DN20

18 qcv > fuel-gen1 DN15

19 qcv > fuel-gen2 DN15

20 fuel-mainP > return DN15

21 fuel-mainS > return DN15

22 return > service DN15

Brown: revised fuel routes. Grey: accepted source services. Pale green: proposed access reservations. Red dots: proposed ports; see terminal schedule.

Sharp corners show clearance envelopes only. Bend/flange/support design remains open. Tank partitions and internal manifold assemblies are not fabricated geometry.

JOHANSSON SHIPYARD | HULL 127 | MINATO MARU

REV C1 | 2026-10-17 | FUEL-ONLY ADDENDUM | A3 | SCALE 1:50

DRAWN: generated | CHECK: internal geometric checks | APPROVAL: NONE

Reference: immutable Revision B (13 October 2026). Proposed interfaces; no supplier or construction approval.

JS127-FUEL-C1-03 | Proposed terminal schedule 1

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

Port / source node	X mm	Y mm	Z mm	Outward	Owner / basis
storeP:fill	12000	2250	680	up	proposed assembly/reservation
storeS:fill	12000	-2250	680	up	proposed assembly/reservation
storeP:suction	12300	2250	680	up	proposed assembly/reservation
storeS:suction	12300	-2250	680	up	proposed assembly/reservation
storeP:overflow	12650	2250	680	up	proposed assembly/reservation
sludge:inlet	13000	2980	680	up	proposed assembly/reservation
bunker:fillP	12850	3660	3400	down	proposed assembly/reservation
bunker:fillS	12850	3840	3400	down	proposed assembly/reservation
vP:tank	12300	2250	1250	down	proposed assembly/reservation
vP:pump1	12300	2160	1350	up	proposed assembly/reservation
vP:pump2	12300	2339	1350	up	proposed assembly/reservation
vS:tank	12300	-2250	1250	down	proposed assembly/reservation
vS:pump1	12300	-2340	1350	up	proposed assembly/reservation
vS:pump2	12300	-2160	1350	up	proposed assembly/reservation
tp1:suction-vP	14200	3630	1450	up	fuel transfer no.1 pump
tp1:suction-vS	14200	3830	1450	up	fuel transfer no.1 pump
tp1:discharge	14240	3480	1340	stbd	fuel transfer no.1 pump
purifier:feed1	12010	3480	1700	stbd	fuel purifier
tp2:suction-vP	14550	3630	1450	up	fuel transfer no.2 pump
tp2:suction-vS	14550	3830	1450	up	fuel transfer no.2 pump
tp2:discharge	14590	3480	1340	stbd	fuel transfer no.2 pump
purifier:feed2	12170	3480	1700	stbd	fuel purifier
purifier:clean	12220	3710	2000	up	fuel purifier
purifier:sludge	11820	3480	1200	stbd	fuel purifier

All coordinates are proposed interfaces. Tank Z=680 mm is the shared reservation roof, not a verified tank roof. Pump/purifier branches require selected-maker manifolds. Bunker station has a common inlet/tee with two isolated outlets; internal assembly is a design hold.

JS127-FUEL-C1-04 | Proposed terminal schedule 2

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

Port / source node	X mm	Y mm	Z mm	Outward	Owner / basis
service:clean-in	13350	-3230	2250	port	fuel service tank
overflow:inlet	12800	-3020	2400	stbd	proposed assembly/reservation
service:overflow	12800	-3230	2400	port	fuel service tank
overflow:outlet	12800	-2940	2400	up	proposed assembly/reservation
service:supply	13600	-3230	1300	port	fuel service tank
qcv:inlet	13600	-3080	1300	stbd	proposed assembly/reservation
qcv:out-mainP	13600	-3080	2400	up	proposed assembly/reservation
fuel-mainP:inlet	10000	1212	2050	up	proposed assembly/reservation
qcv:out-mainS	13600	-2980	2400	up	proposed assembly/reservation
fuel-mainS:inlet	10000	-1212	2050	up	proposed assembly/reservation
qcv:out-gen1	13600	-2880	2400	up	proposed assembly/reservation
fuel-gen1:inlet	15300	750	2000	up	proposed assembly/reservation
qcv:out-gen2	13600	-2780	2400	up	proposed assembly/reservation
fuel-gen2:inlet	15300	-750	2000	up	proposed assembly/reservation
fuel-mainP:return	10180	1212	2050	up	proposed assembly/reservation
return:in-mainP	13000	2800	2400	up	proposed assembly/reservation
fuel-mainS:return	10180	-1212	2050	up	proposed assembly/reservation
return:in-mainS	13000	3050	2400	up	proposed assembly/reservation
return:outlet	13150	2930	2400	up	proposed assembly/reservation
service:return	13050	-3230	2300	port	fuel service tank

All coordinates are proposed interfaces. Tank Z=680 mm is the shared reservation roof, not a verified tank roof. Pump/purifier branches require selected-maker manifolds. Bunker station has a common inlet/tee with two isolated outlets; internal assembly is a design hold.

JS127-FUEL-C1-05 | Connection disposition and sizing basis

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

Connection	Rev B issue	C1 external geometry	Length m	Pipe gap mm
bunker>storeP#0	missing physical terminal	geometry-clear	4.98	23
bunker>storeS#1	missing physical terminal	geometry-clear	9.66	32
storeP>vP#2	geometry-clear	geometry-clear	0.57	95.869
storeS>vS#3	geometry-clear	geometry-clear	0.57	96.536
vP>tp1#4	geometry-clear	geometry-clear	5.872	14.202
vS>tp2#5	geometry-clear	geometry-clear	10.782	14
vP>tp2#6	pipe clearance	geometry-clear	6.565	20
vS>tp1#7	pipe clearance	geometry-clear	12.852	20
tp1>purifier#8	pipe clearance	geometry-clear	5.15	11
tp2>purifier#9	pipe clearance	geometry-clear	5.438	11
purifier>service#10	geometry-clear	geometry-clear	9.182	18
purifier>sludge#11	pipe clearance	geometry-clear	2.2	65
service>overflow#12	geometry-clear	geometry-clear	0.21	33.6
overflow>storeP#13	pipe clearance	geometry-clear	7.6104	33.6
service>qcv#14	pipe clearance	geometry-clear	0.15	490.833
qcv>fuel-mainP#15	geometry-clear	geometry-clear	8.7785	20
qcv>fuel-mainS#16	geometry-clear	geometry-clear	6.0535	20
qcv>fuel-gen1#17	geometry-clear	geometry-clear	6.43	47
qcv>fuel-gen2#18	geometry-clear	geometry-clear	13.31	14
fuel-mainP>return#19	pipe clearance	geometry-clear	4.9975	24
fuel-mainS>return#20	pipe clearance	geometry-clear	8.0325	53
return>service#21	pipe clearance	geometry-clear	7.61	11.31

DN retained. At nominal ID=25 mm only: 2 m3/h implies 1.1318 m/s; 0.25 m3/h implies 0.1415 m/s. These arithmetic screens are not a pipe class or hydraulic sizing approval. Bunker/return/overflow/sludge design flows and actual IDs are unavailable.

JS127-FUEL-C1-06 | Scope, open holds and revision control

FOR DESIGN REVIEW - CONSTRUCTION NOT APPROVED

1. Select equipment and confirm nozzle sizes, directions, flange envelopes and pump suction/discharge manifolds.

2. Design individual storage/sludge tank partitions, roof plating, penetrations, dip pipes, vents, sounding and overflow arrangements; the source defines only a shared tank reservation.

3. Select pipe class, actual ID/OD, wall, corrosion allowance, insulation, bend radius, flange/valve access and supports; current radii are geometric reservations.

4. Confirm pump curves, viscosity/temperature, NPSH, pressure losses, purifier feed regulation/bypass and return/overflow capacity. A 2 m3/h pump is not demonstrated compatible with a 0.25 m3/h purifier.

5. Confirm gravity head at minimum service-tank level, engine inlet/return pressure and generator return arrangements. The old 1.8 m gravity-head claim is unsupported.

6. Engineer and approve tank/deck/watertight bulkhead sleeves, fire protection and fuel containment. Proposed penetrations confer no structural or fire approval.

7. Coordinate unvalidated service routes, cables, ventilation, removal and escape envelopes; this check only reserves stated access boxes and existing accepted pipes.

Fuel external routes: 22 geometry-clear, 0 open; 12 prior geometric issues resolved.

Whole vessel register: 67 geometry-clear, 139 unresolved positive-bore connections.

Original Revision B bytes are unchanged. C1 supersedes fuel routing/interfaces only; all other Revision B holds remain.

Validation is internal, against a fictional source model. No flag, class, surveyor or professional engineering certification is claimed.